Answer **all** the questions in this Section in the spaces provided.

1 A uniform rod of mass 0.200 kg and length 0.90 m is suspended at one end by a light, inextensible string of length 0.37 m, while its other end rests on a rough horizontal surface, as shown in Fig. 1.1.

The rod is inclined at an angle of 40° to the horizontal. The string is inclined at an angle θ to the rod, where $\theta = 62°$. The upper end of the string is attached to a fixed point vertically above the midpoint of the uniform rod.

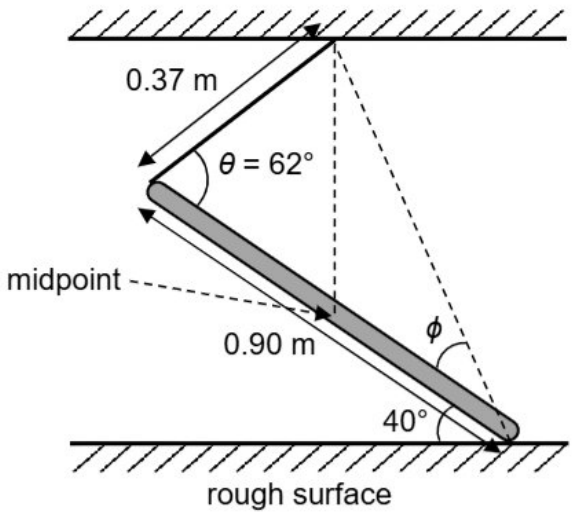


Fig. 1.1. (not drawn to scale)

(a) State the *principle of moments*.

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..... [1]

(b) Draw a free-body diagram of the uniform rod, indicating all forces acting on it, including the reaction force by the rough surface.

[1]

(c) By determining the angle ϕ, show that the force exerted by the rough surface on the rod acts at an angle of 64° to the horizontal.

[2]

(d) Hence, determine the magnitude of the force exerted by the rough surface on the rod.

force = ... N [2]

(e) Explain why the angle of 64° in **(c)** would remain the same even when an additional mass is hung from the midpoint of the rod. Assume that the rod does not slide even when the additional mass is hung.

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..... [2]

(f) State and explain how the angle θ would change if the horizontal surface were frictionless.

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..... [2]